Eg:- Stock A : 2, 3, 5, 4, 6

Stock B : 5, 8, 10, 11, 14

Step1: Calculate the Mean of each Stock

Find the average price of each stock.

$$\text{Mean of A} = \frac{\Sigma x}{n} = \frac{2+3+5+4+6}{5} = 4 \ (\bar{x})$$

$$\text{Mean of B} = \frac{\Sigma B}{n} = \frac{5+8+10+11+14}{5} = 9.6 \ (\bar{y})$$

Step2: Find Deviation from the Mean:

For A: $(x - \bar{x}) = (-2, -1, 1, 0, 2)$

For B = $(y - \bar{y}) = (-4.6, -1.6, 0.4, 1.4, 4.4)$

Step3: Multiply corresponding Deviation:-

$\Sigma (x - \bar{x})(y - \bar{y})$

$(-2) \times (-4.6) = 9.2$; $(-1) \times (-1.6) = 1.6$, $1 \times 0.4 = 0.4$,

$0 \times 1.4 = 0$, $2 \times 4.4 = 8.8$

Step4: Sum of the product.

$$\text{Sum} = 9.2 + 1.6 + 0.4 + 0 + 8.8 = 20$$

Step5: Calculate Covariance

Divide the sum by the No. of observation minus one $(n-1)$

$$\text{covariance} = \frac{\sum_{i=1}^{n} (x - \bar{x})(y - \bar{y})}{N-1} = \frac{20}{4} = 5$$

$\text{Cov}_{(x,y)}$

Interpretation:-

A positive covariance indicate that the stocks tend to move in the same direction, while a negative covariance suggest they move in opposite directions (1, 3, 5), ,

In this case, the covariance is positive, indicating that stocks A and B tend to ~~raise~~ rise or fall togeather.

8-1

Covariance Matrix

x (2, 4, 6, 8, 10), y (3, 7, 10, 14, 17)

Sample Covariance

$$Cov(x,y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{N-1}$$

Population Covariance

$$cov(x,y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{N}$$

x	y	$x - \bar{x}$	$y - \bar{y}$	$(x-\bar{x})(y-\bar{y})$
2	3	-4	-7.2	28.8
4	7	-2	-3.2	6.4
6	10	0	-0.2	0
8	14	2	+3.8	7.6
10	17	4	6.8	27.2
Σ30	51	0	0	70

$$\bar{x} = \frac{\sum x}{n} = \frac{30}{5} = 6$$

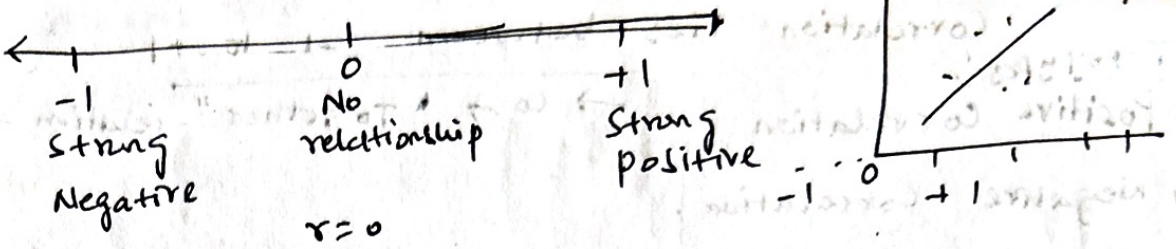
$$\bar{y} = \frac{\sum y}{n} = \frac{51}{5} = 10.2$$

$$Cov_{(x,y)} = \frac{\sum (x_i - \bar{x})(y - \bar{y})}{N-1} = \frac{70}{5-1} = \frac{70}{4} = \boxed{17.5}$$

Interpretation:

Correlation Co-efficient:

The coefficient of Correlation (r) is a Measure of the Strength of the Linear relationship between two variables.



Limitation of correlation:

→ correlation does not tell us: Everything about the data. Mean and standard deviation continues to be importants.

→ the data may be described by a curve more complicated than a straight line, but this will not show up in Calculation of r.

Example: E-commerce Business website Vs Money spent by a Customer

Amazon!
Flipkart

Number of unique user Vs sales in a day

Measure the correlation Between the patient's blood pressure and the medication used.

Pg 1

What is correlation?

Correlation is a statistical techniques used to determind the degree to which two variables are related.

Ex: the price and demand of a product is in correlation.

Correlation lies between -1 to +1

→ co → "Together" - relation - "connection"

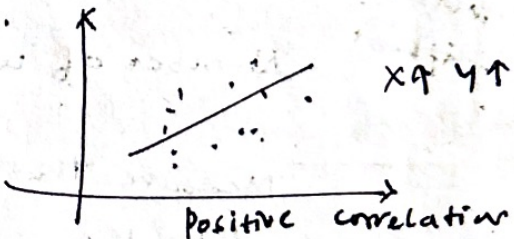
Two Types:
→ Positive Correlation
→ Negative Correlation

Positive Correlation:

Def: It's relationship between two variables where if one variable increases, the other one also increases and vise-versa.

→ the linear relationship r/p can be positive,
Two variable can Increases / Decreases in the same direction.

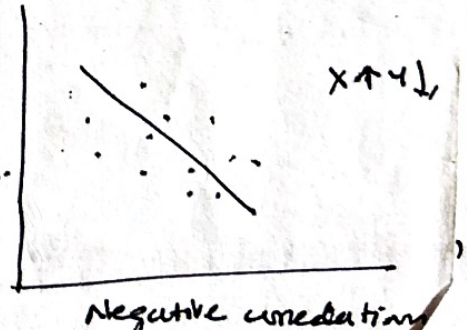
→ As the number of trees cut down increase, the probability of erosion increases.



Negative Correlation:

→ Negative correlation means there is an inverse relationship between two variable. when one decreases, other increases.

Ex: If a car decreases speed, the time taken to reach the destination increases.



Linear Regression

↳ To perform a predictive Analysis. It's relationship Between two variables.

→ which means that changes in the independent variables are associated with proportional changes in the dependent variable

LR: $y = a + bx$

$$a = \frac{[(\Sigma y)(\Sigma x^2) - (\Sigma x)(\Sigma xy)]}{[n(\Sigma x^2) - (\Sigma x)^2]}$$

$$b = \frac{[n(\Sigma xy) - (\Sigma x)(\Sigma y)]}{[n(\Sigma x^2) - (\Sigma x)^2]}$$

Least Square Method:- It's most common method used to fit a regression line in the x-y graph.

→ line of best fit by reducing the sum of the square of the vertical deviation from each data point to the line.

→ Any point that is fitted accurately. It's perpendicular deviation is zero

$$y = B_0 + B_1 X$$

$$B_1 = b_1 = \Sigma [(x_i - x)(y_i - y)] / \Sigma [(x_i - x)^2]$$

Error in LR

↳ Standard error → Measure the Average proportion

x	y	x^2	xy
3	8	9	24
9	6	81	54
5	4	25	20
3	2	9	6

$\Sigma x = 20/4 = 5$ $\Sigma y = 20/4 = 5$ $\Sigma x^2 = 124/4 = 31$ $\Sigma xy = 104/4 = 26$

$$a = \frac{\Sigma y \cdot \Sigma x^2 - \Sigma x \Sigma xy}{n(\Sigma x)^2 - (\Sigma x)^2}$$

$$a = \frac{\{20(124) - 20(104)\}}{\{4(124) - 400\}} = 400/96 = 4.17$$

$$b = \frac{n\Sigma xy - (\Sigma x)(\Sigma y)}{n\Sigma x^2 - (\Sigma x)^2}$$

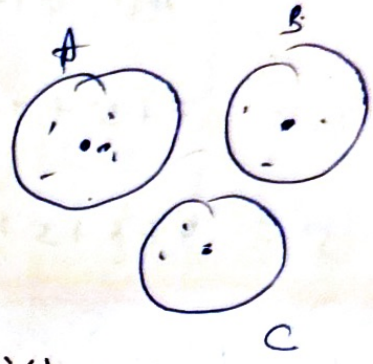
$$b = \frac{\{4(104) - 20(20)\}}{\{4(124) - 400\}} = 16/96 = 0.166$$

↳ Equation = $y = a + bx \rightarrow$ $y = 4.17 + 0.166x$

P-1

K-Means clustering - problem Statement

↳ Suppose that the Data Mining task is to cluster points into into Three clusters.



→ where the points are

→ A1 (2, 10), A2 (2, 5), A3 (8, 4)
B1 (5, 8), B2 (7, 5), B3 (6, 4)
C1 (1, 2), C2 (4, 9), C3.

→ the Distance function is Euclidean distance.

$$d(P_1, P_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

→ Suppose initially we assign A1, B1 and C1 as the center of each cluster respectively.

Step 1:

Initial centroids:

A1: (2, 10)
B1: (5, 8)
C1: (1, 2)

Data point	x_i	y_i	Distance to C1 (2, 10)	Distance to C2 (5, 8)	Distance to C3 (1, 2)	Cluster	New cluster
A1	2	10	0	3.61	8.06	1	
A2	2	5	5	4.24	3.16	3	
A3	8	4	8.49	5.0	7.28	2	
B1	5	8	3.61	0.0	7.21	2	
B2	7	5	7.07	3.61	6.71	2	
B3	6	4	7.21	4.12	5.39	2	
C1	1	2	8.06	7.21	0.0	3	
C2	4	9	2.24	1.41	7.62	2	

ED = $d(P_1, P_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

A1 = $\sqrt{(2-2)^2 + (10-10)^2}$

$(x_1, y_1) = (2, 10)$ A1 = 0

centroid = $(x_2, y_2) = (2, 10)$ A2 = $\sqrt{(2-2)^2 + (10-5)^2}$

A1 A1 = $\sqrt{25} = 5$

New Centroids:

Cluster : 1

$A_1 - 2, 10 \div 1 = (2, 10)$

cluster 2: $A_3 + B_1 + B_2 + B_3 + C_2 / 5$

$x = 8+5+7+6+4 / 5 = 6$

$y = 4+6+5+4+9 / 5 = 6$

$B_1 : (6, 6)$,

Cluster 3

C_1: $A_2 + C_1 / 2$ x $2 + 1 / 2 = 3/2$

y $5 + 2 / 2 = 7/2$

$C_1 = (1.5, 3.5)$

New Centeroids:

$A_1 : (2, 10)$

$B_1 : (6, 6)$

$C_1 : (1.5, 3.5)$

Current Centroids:

$A_1 : (2, 10)$

$B_1 : (6, 6)$

$C_1 : (1.5, 3.5)$

Data Points			Distance to (2, 10)	Distance to (6, 6)	Distance to (1.5, 3.5)	Cluster	New cluster
A_1	2	10	0.0	5.66	6.52	1	1
A_2	2	5	5.0	4.12	1.58	3	3
A_3	8	4	8.49	2.83	6.52	2	2
B_1	5	8	3.61	2.24	5.70	2	2
B_2	7	5	7.07	1.41	5.70	2	2
B_3	6	4	7.21	2.0	4.53	2	2
C_1	1	2	8.06	6.40	1.58	3	3
C_2	4	9	2.24	3.61	6.04	(1)	(1)

New centroids:

A_1: $A_1 + C_2 / 2 = 2+4 / 2 = 3$

$10+9 / 2 = 9.5$

$(3, 9.5)$

B_1: $A_3 + B_1 + B_2 + B_3 / 4$ $\therefore 8+5+7+6 / 4 = 6.5$

$4+8+5+4 / 4 = 5.25$

$(6.5, 5.25)$

C_1: $A_2 + C_1 / 2 =$ $\therefore 2+1 / 2 = 1.5$

$5+2 / 2 = 3.5$

$(1.5, 3.5)$

Current centroids:

A1: (3, 9.5)

B1: (6.5, 5.25)

C1: (1.5, 3.5)

Datapoints			Distance to				cluster	New cluster
			3 / 9.5	6.5 / 5.25	1.5 / 3.5			
A1	2	10	1.12	6.54	6.52		1	1
A2	2	5	4.61	4.51	1.58		3	3
B3	8	4	7.43	1.95	6.52		2	2
B1	5	8	2.50	2.13	5.70		2 → 1	
B2	7	5	6.02	0.56	5.70		2	2
B3	6	4	6.26	1.35	4.53		2	2
C1	1	2	7.76	6.39	1.58		3	3
C2	4	9	1.12	4.51	6.04		1	1

New centroids:

A1: A1 + B1 + C2 / 3

(3.66, 9)

B1: A3 + B2 + B3 / 3

(7, 4.33)

C1: A2 + C1 / 2

(1.5, 3.5)

11/3

2 + 5 + 4 / 3 = 3.66

10 + 8 + 9 / 3 = 27/3 = 9

8 + 7 + 6 / 3 = 21/3 = 7

4 + 5 + 4 / 3 = 13/3 = 4.33

2 + 1 / 2 = 1.5

5 + 2 / 2 = 3.5

Current centroids:

A1: (3.67, 9)

B1: (7, 4.33)

C1: (1.5, 3.5)

Datapoints			Distance to				cluster	New cluster
			3.67 / 9	7 / 4.33	1.5 / 3.5			
A1	2	10	1.94	7.56	6.52		1	1
A2	2	5	4.33	5.04	1.58		3	3
A3	8	4	6.62	1.05	6.52		2	2
B1	5	8	1.67	4.18	5.70		1	1
B2	7	5	5.21	0.67	5.70		2	2
B3	6	4	5.32	1.05	4.53		2	2
C1	1	2	7.49	6.44	1.58		3	3
C2	4	9	0.33	5.55	6.04		1	1

same.

Final cluster formation:

A1: A1, B1, C2,

B2: A3, B2, B3, C1

C1: A2, C1

} Final Centroid values of Each clusters of k-means Algorithm.

K- Medoid clustering Algorithms:-

→ Apply k-medoid clustering Algorithm to form two clusters
→ use Manhattan distance to find the Between Datapoint and Medoid

Step 1:

→ select Two Medoids :-

$c_1 = (3,4)$, $c_2 = (7,4)$

(x_1, y_1) and (x_2, y_2) are data points

Manhattan dist $= |x_1 - x_2| + |y_1 - y_2|$

cordinate of $x_1 - x_2$

Mdist $[(2,6), (3,4)] = |2-3| + |6-4| = 3$

Mdist $[(3,4), (3,4)] = |3-3| + |4-4| = 0$

∴ etc

i	x	y	c_1	c_2	cluster
x_1	2	6	3	7	c_1
x_2	3	4	0	4	c_1
x_3	3	8	4	8	c_1
x_4	4	7	4	6	c_1
x_5	6	2	5	3	c_2
x_6	6	4	3	1	c_2
x_7	7	3	5	1	c_2
x_8	7	4	4	0	c_2
x_9	8	5	6	2	c_2
x_{10}	7	6	6	2	c_2

Step 2:

cluster are :

c_1 : { (2,6), (3,4), (3,8), (4,7) }

c_2 : { (6,2), (6,4), (7,3), (7,4), (8,5), (7,6) }

Total cost of cluster : (3,4) compare to Individual cost (2,6)

" (3,8)

(3,4) " (4,7)

(3,4 "

Calculate the Total cost

$$cost(c, x) = \sum_i |c_i - x_i|$$

Total cost = { cost((3,4),(2,6)) + cost((3,4),(3,8)) +
cost((3,4),(4,7)) + cost((7,4),(6,2)) + cost((7,4),(6,4)) +
cost((7,4),(7,3)) + cost((7,4),(8,5)) + cost((7,4),(7,6)) }

= cost(3-2, 4-6), cost[(3-3)-(4-8)| ....
1 + 2 = 3 | 0 + (-4)| = 4

Total cost = 3+4+4+3+1+1+2+2 = 20.

Step 3: → Randomly select one non-medoid point and recalculate the cost
→ Swap c_2 with O (x7) O=(7,3)
→ New Medoids: {Existing Medoids, New}
→ $c_1 = (3,4)$ and O = (7,3)

Manhattan Dist = $|x_1 - x_2|$ + $|y_1 - y_2|$

Dist [(2,6),(7,3)] = |2-7| + |6-3| = 8
[(3,4),(7,3)] = [3-7| + |4-3| = 5
4 + 1

i	x	y	c_1	O	current
x1	2	6	3	8	c_1
x2	3	4	0	5	c_1
x3	3	8	4	9	c_1
x4	4	7	4	7	c_1
x5	6	2	5	2	0
x6	6	4	3	2	0
x7	7	3	5	0	0
x8	7	4	4	1	0
x9	8	5	6	3	0
x10	7	6	6	3	0

New clusters:

$c_1 : \{(2,6), (3,4), (3,8), (4,7)\}$

$O : \{(6,2), (6,4), (7,3), (7,4), (8,5), (7,6)\}$

Calculate the Total cost:

$$\text{Cost}(c, x) = \sum_i |c_i - x_i|$$

Current Total cost : { Cost((3,4), (2,6)) + Cost.

Current Total Cost : 3+4+4+2+2+1+3+3 = 22

step 4:

Cost of swapping of medoid c_2 with O

S = Current Total cost − previous Total cost

S = 22 − 20 = 2 > 0 { Current Total cost is more than existing one. So we swap the new medoid value.

- Hence swapping c_2 with O is not a good idea

Final medoids are $c_1 = (3,4)$ and $c_2 = (7,4)$

Hierarchical clusters using a single link Technique.

↳ Agglomerative Method ✓ solved

Problem Definition:

For the given dataset find the clusters using single Link Technique. Use Euclidean distance and draw the Dendrogram.

Soln.

sample No	x	y
P1	0.40	0.53
P2	0.22	0.38
P3	0.35	0.32
P4	0.26	0.19
P5	0.08	0.41
P6	0.45	0.30

Step 1: Compute the distance Matrix

→ So we have to find the Euclidean distance b/w each and every points

→ Let A(x_1, y_1) and B(x_2, y_2) are two points

→ The Euclidean distance Between

$$\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

$$d(P_1, P_2) = \sqrt{(0.22-0.40)^2 + (0.38-0.53)^2}$$

$$= 0.23$$

$$d(P_1, P_3) = \sqrt{(0.35-0.40)^2 + (0.32-0.53)^2}$$

$$= 0.22$$

$$d(P_2, P_3) = \sqrt{(0.35-0.22)^2 + (0.32-0.38)^2}$$

$$= 0.14$$

	P1	P2	P3	P4	P5	P6
P1	0					
P2	0.23	0				
P3	0.22	0.14	0			
P4	0.37	0.19	0.13	0		
P5	0.34	0.14	0.28	0.23	0	
P6	0.24	0.24	0.10	0.22	0.39	0

rise or fall ... indicating that ...

step 2: Merging the Two closest Members.

* Here the minimum value is 0.10 and hence we combine P_3 and P_6. (as 0.10 came in the P_6 rows and P_3 column).
* Now, form cluster of elements corresponding to the minimum value and update the distance matrix.

	P_1	P_2	P_3	P_4	P_5	P_6
P_1	0					
P_2	0.22	0				
P_3	0.22	0.14	0			
P_4	0.37	0.19	0.13	0		
P_5	0.34	0.14	0.28	0.23	0	
P_6	0.24	0.24	0.10	0.22	0.39	0

(6)

	P_1	P_2	P_3, P_6	P_4	P_5
P_1	0				
P_2	0.23	0			
P_3, P_6	0.22	0.14	0		
P_4	0.37	0.19	0.13	0	
P_5	0.34	0.14	0.28	0.23	0

(5×5)

(P_3, P_6)

Now we will update the distance Matrix.

	P_1	P_2	P_3, P_6	P_4	P_5
P_1	0				
P_2	0.23	0			
P_3, P_6	0.22	0.14	0		
P_4	0.37	0.19	0.13	0	
P_5	0.34	0.14	0.28	0.23	0

	P_1	P_2	P_3, P_6, P_4	P_5
P_1	0			
P_2	0.23	0		
P_3, P_6, P_4	0.22	0.14	0	
P_5	0.34	0.14	0.28	0

(M)

4×4

$\{(P_3, P_6), P_4\}$ and (P_2, P_5)

	P_1	P_2	P_3, P_6, P_4	P_5
P_1	0			
P_2	0.23	0		
P_3, P_6, P_4	0.22	0.14	0	
P_5	0.34	0.14	0.28	0

	P_1	P_2, P_5	P_3, P_6, P_4
P_1	0		
P_2, P_5	0.23	0	
P_3, P_6, P_4	0.22	0.14	0

3×3

the covariance is positive, indicating that stocks

Single link

Sol

$[\{(P3, P6), P4\}, (P2, P5)]$

	P1	P2, P5	P3, P6, P4
P1	0		
P2, P5	0.23	0	
P3, P6, P4	0.22	0.14	0

3×3

	P1	P2, P5, P3, P6, P4
P1	0	
P2, P5, P3, P6, P4	0.22	0

2×2

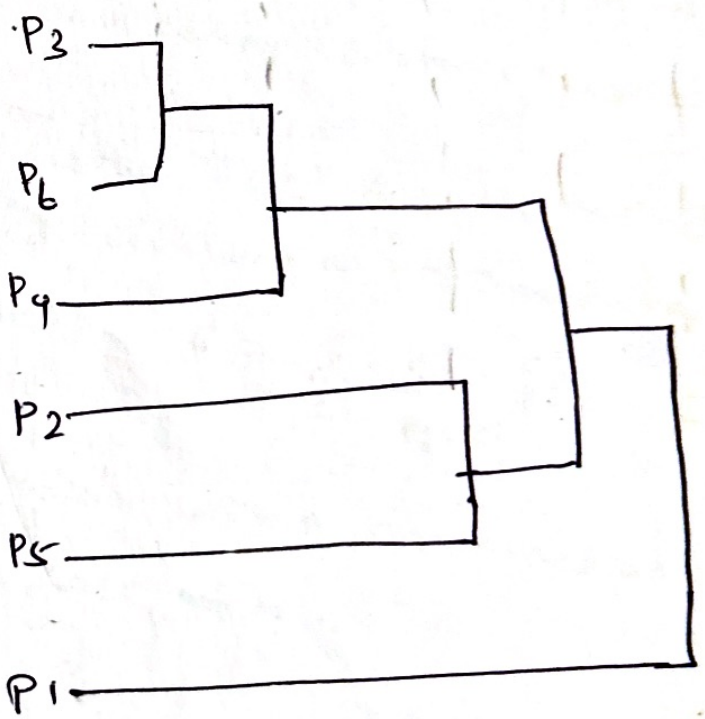
$[\{(P3, P6), P4\}, (P2, P5)]$

$[\{(P3, P6), P4\}, (P2, P5)]$ P1

Note

So now we have reached the solution, the dendrogram for the question will be followed

$[\{(P3, P6), P4\}, (P2, P5)], P1$



Dendogram of the Cluster formed

K-medoid. Vs k-means clustering

→ Partitioning Around Medoids or the k-medoids algorithm is a Partional Clustering Algorithm which is slightly modified from the k-mean Algorithms.

→ In k-means Alg, they choose means as the Centroid but in the k-Medoids, data points are chosen to be the medoids.

K-medoid cluster Algorithms

1. Initially select k- random point as the Medoids from the given n-data points of the data set.
2. Associate each data point to the closest medoid by using any of the most common distance metries.
3. Calculate the cost as the Total Sum of the distance ~~of the~~ (also called dissimilarities) of the data points from the assigned medoid. $C = \sum_{ci} \sum_{P_i \in ci} |P_i - K_i|$
4. Swap one medoid point with a non-medoid point and re calculate the cost. ~~of it~~.
5. If the calculated (Current) cost with the new medoid point is more than the previous cost (function), we undo the swap, and the Algorithm coverages else, we repeat step 4.

Adv. k-medoids Algorithm

→ It's simple to understand and easy to implement.

→ k-Medoid Algorithm is fast and coverage in a fixed number of steps.

→ K-medoid Algorithm is less sensitive to outliers than other partitioning algorithms.

Dis.Adv

→ The main DisAdvantage of k-medoid Alg is that it's not suitable for clustering non-spherical (Arbitrarily shaped) group of objects.

→ It may obtain different results for different runs on the same dataset because the first k-medoids are chosen randomly.